Minimum Description Length Principle

With An Application For Credit Risk Models

Paul van Leeuwen (paul.vanleeuwen@devolksbank.nl)

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Introduction

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MDL in Practice

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Introduction

The Speaker

- Paul van Leeuwen, employee of KRT RDS 5.0 as of September 2022.
- Up to 2017 part of the Modelling team at dVB.
- After that as Model Validator at Achmea and Lead Data Scientist at Wageningen University & Research.
- Now self-employed and currently working in the financial sector and providing `R` workshops.

Why this subject?

- Improve current approaches and modelling techniques.
- Stay in the forefront of statistical innovation.
- Connection with other realms of statistics.

What is MDL?

Introduction

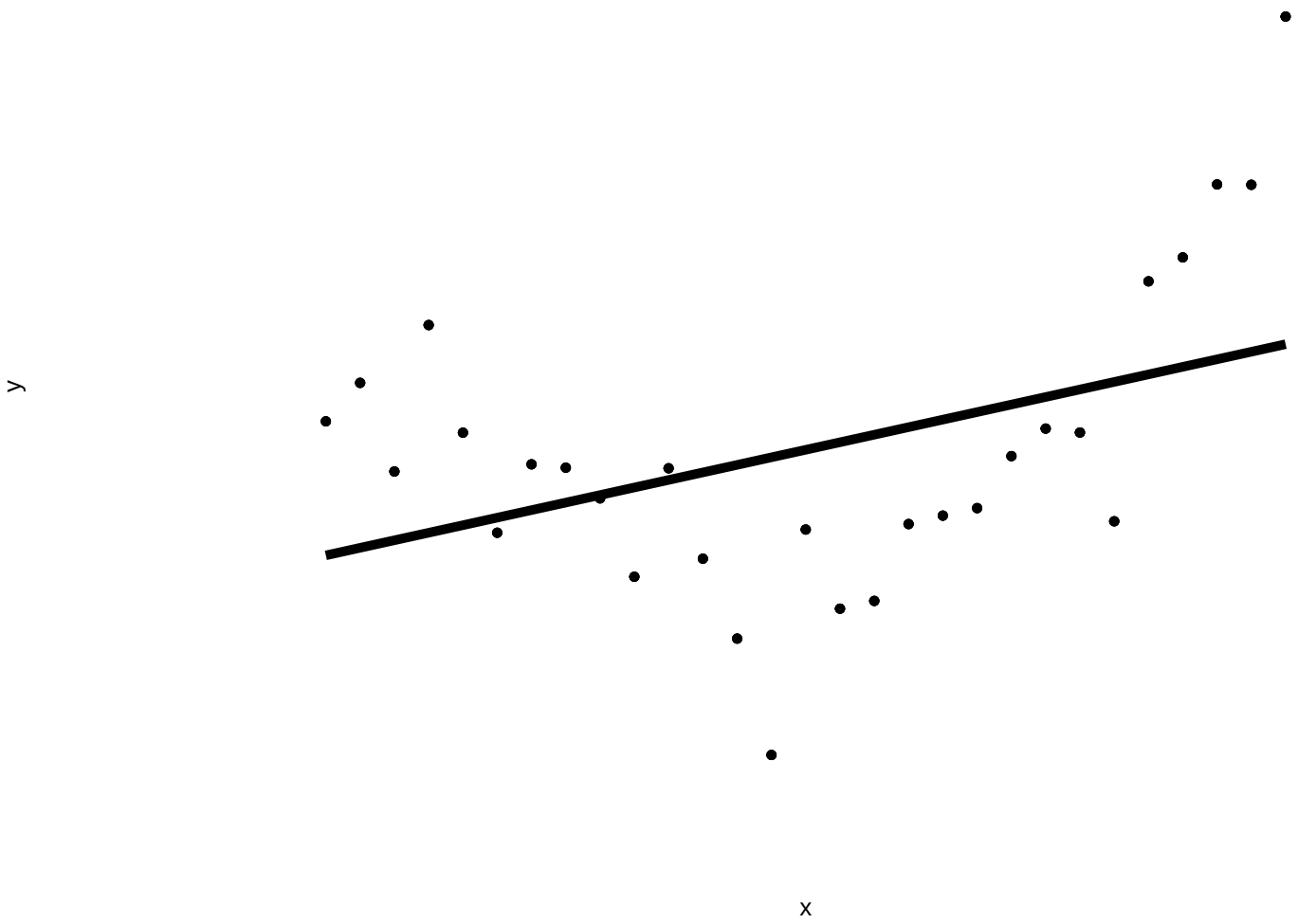
- The Minimum Description Length (MDL) principle aims to describe the data and its description mechanism with the smallest possible information 'length'.
- Applications (among else):
  - model selection (e.g. what order of the Markov model family do we want?);
  - deal with overfitting (e.g. how many explanatory variables to include);
  - exploratory data analysis (what prior knowledge can we confirm?).
- Close ties with frequentist statistics, Bayesian statistics, and machine learning.
- Why is MDL relatively unknown?
  - MDL is the intersection of advanced measure theory, information theory, and statistics.
  - For a decent introduction into MDL, see (Grünwald 2007).

Example of dealing with overfitting

What polynomial generated by this dataset?

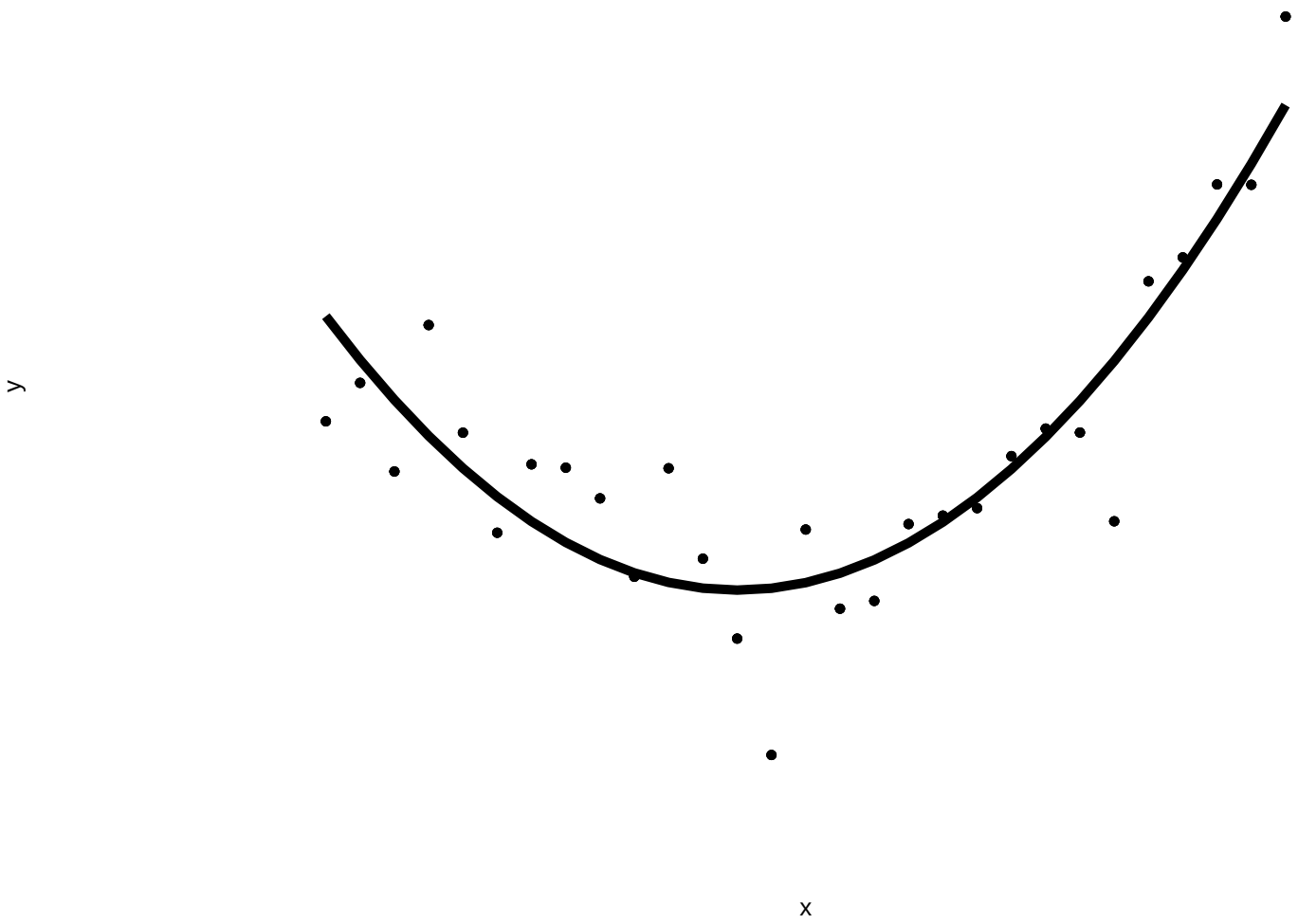
Example of dealing with overfitting

linear fit $\hat{y} = \beta_0 + \beta_1 x$



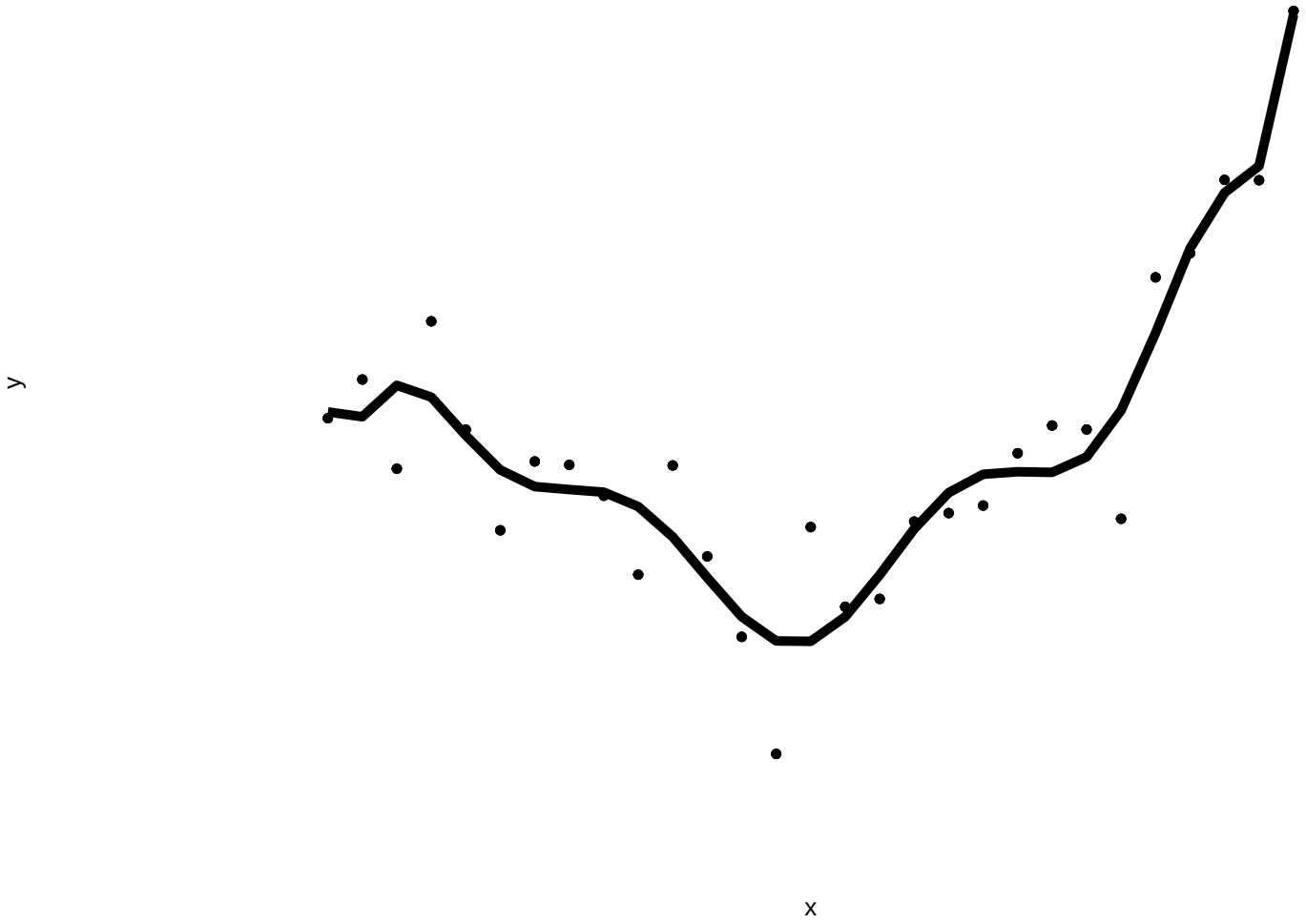
Example of dealing with overfitting

quadratric fit $\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$



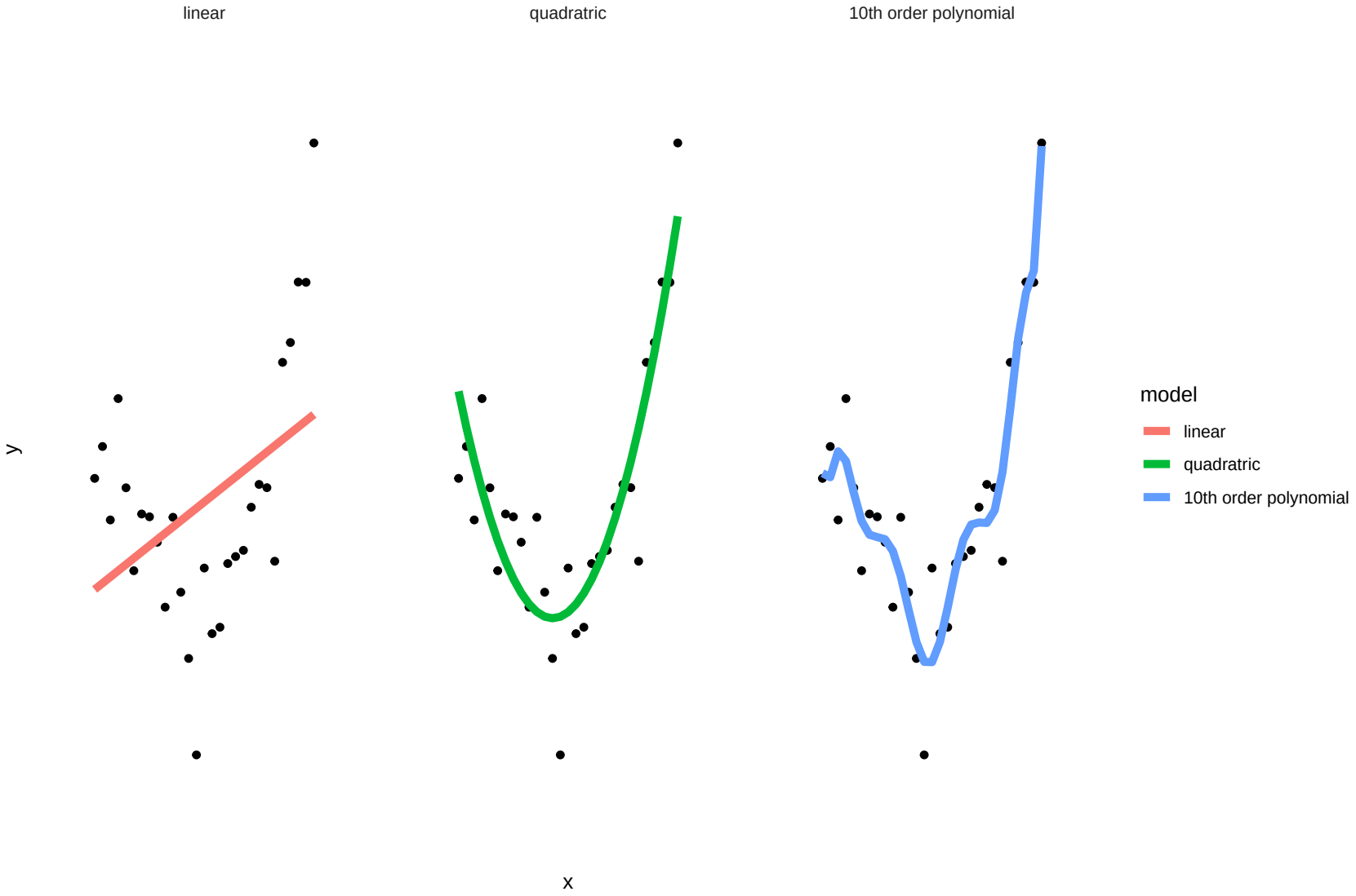
Example of dealing with overfitting

10th order polynomial fit $\hat{y} = \beta_0 + \beta_1 x + \ldots + \beta_{10} x^{10}$



Example of dealing with overfitting

data generated by $x^3 + 2x^2 + 5 + \varepsilon$, $\varepsilon \sim N(0, 1)$



How does MDL work?

- Patterns or regularities in the data can be described with less information 'length' compared to the data alone.
- less information 'length' = compression
- Choose the model that gives the shortest description of the data.
- Note that MDL is an approach, not an algorithm.
  - The modeller has to make choices to implement the MDL principle.

Example of MDL

- Consider three data-generating processes (dgp's) that generate each a binary sequence of length 1000:
  1. 1000100010001000100010001000100010001000100 ... 1000
  2. 1001111110111100011001110001010100110001101 ... 1010
  3. 0100010000001011100100000000001101000110000 ... 0100

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- All sequences written out take 1000 bits to be reproduced.
- However, because of regularities present in the dgp's, we require less bits to reproduce the same sequences.
- Question: how many bits of Matlab code does each sequence take to be *exactly* reproduced?

Example of MDL

- The sequences as before:
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Example of MDL

- The sequences as before:
  1. 1000100010001000100010001000100010001000100 ... 1000:
     repetition of `[1 0 0 0]` 250 times.
     Matlab-code:
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     textDgp_1 = 'repmat([1 0 0 0],1,250)'
     whos_textDgp_1 = whos('textDgp_1')
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     yields 23 bytes = 184 bits, a compression ratio of
     $1 - \frac{184}{1000} = 81.6\%$!

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  3. 0100010000001011100100000000001101000110000 ... 0100:
     a roll with a four-sided die with outcomes 2, 3, 4 assigned to 0 and outcome 1 assigned to 1; no compression possible because of the randomness involved.

The Principle of MDL

- The more randomness involved, the less data compression is possible.
- When the restriction of exact reproduction is alleviated we may obtain some data compression.
- Dgp 2 (the coin toss) implies practically no possible data compression.
  - As the data sequence is completely random.
- Dgp 3 (the 4-sided die) implies some possible data compression.
  - Although exact reproduction is not possible the set of this type of data sequences requires around 821 bits.
- Most datasets are almost incompressible.
  - Only a small fraction can be significantly compressed.

Description Methods

- A description method in conjunction with a coding scheme maps an object to a number of bits.
  - The coding alphabet $\mathbb{B}$ can be binary ($\mathbb{B} = \{0, 1\}$), the Western alphabet ($\mathbb{B} = \{\mathrm{a}, \mathrm{b}, \ldots, \mathrm{z}\}$), etc.
  - The mapping is a one-many relation and many-one.
  - Mathematically: a dataset $D = (x_1, \ldots, x_n)$ with $x_i \in \mathbb{B}$ from a sample space $\mathcal{X}^n$ is mapped to $\mathbb{R}_+$ by $L\colon \mathcal{X}^n \mapsto \mathbb{R}_+$.
- Any alphabet is fine.
- For example, to map an integer from $\{1, \ldots, n\}$ in a uniform way, we need $\log_2 n$ bits.
  - Note that n has to be known in advance.
  - For example, take $n = 64$. Then we have 64 binary data sequences of length $\log_2(64) = 6$.

The Principle of MDL (continued)

- To describe any dataset we need $L(D, H)$ bits.
  - Both the description method H and the data D require storage space.
- The MDL principle employed for model selection is to minimise the sum of
  - the number of bits to encode the description mechanism $L(H)$ and
  - the number of bits to encode, with the description mechanism H, the data observed $L(D|H)$.
- Information 'length' is this sum $L(H) + L(D|H)$.
- The former is the *model complexity*, the latter the *fit of the data*.
- The MDL principle is to choose the model specification as to minimise this sum.

The Principle of MDL (continued)

- More mathematically, given a set of candidate models $\mathcal{H} = \{\mathcal{H}_1, \mathcal{H}_2, \ldots\}$, select the optimal model $\mathcal{H}^\star \in \mathcal{H}$ as

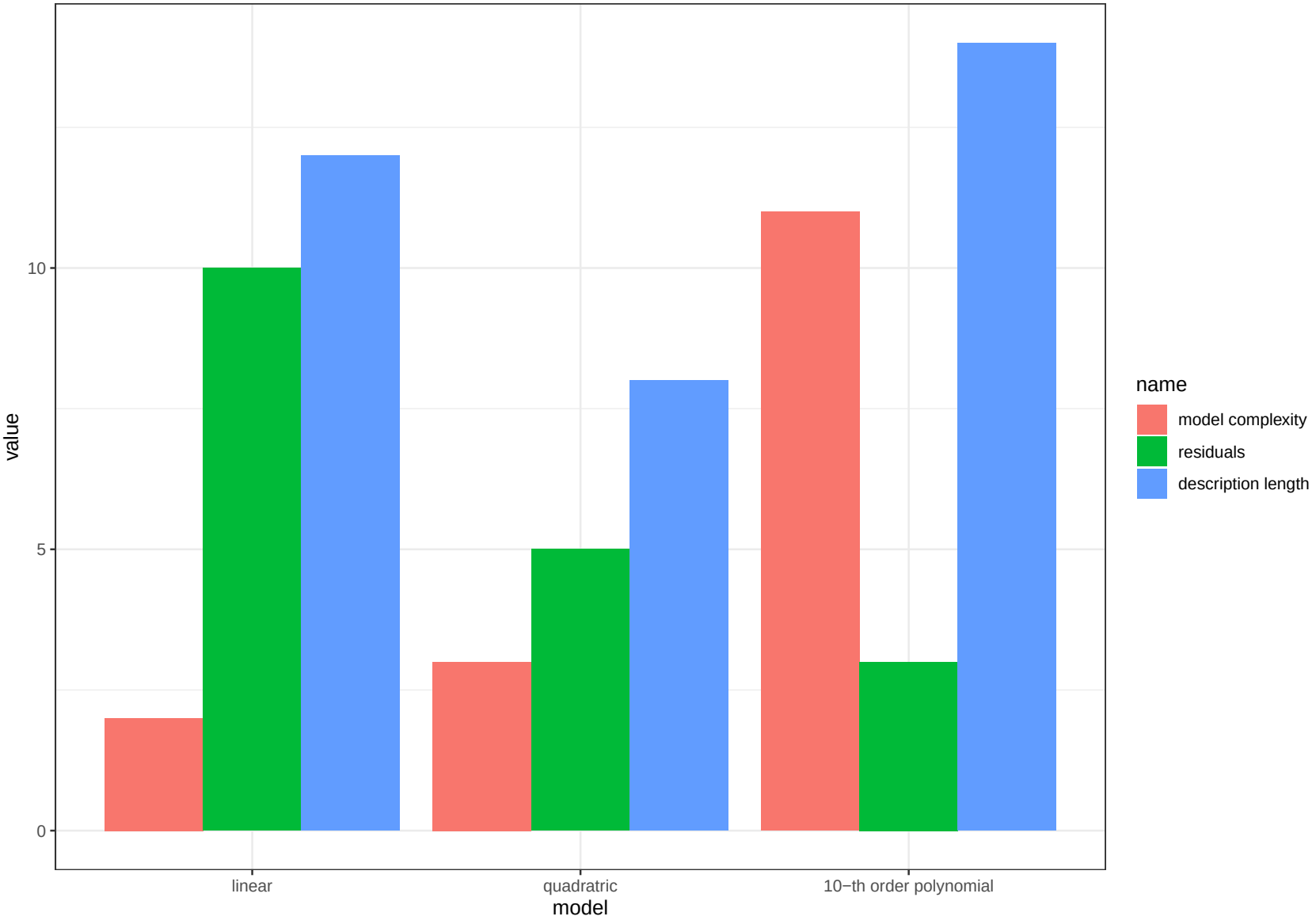
$$\mathcal{H}^\star := \arg\min_{H \in \mathcal{H}} L(D, H) = \arg\min_{H \in \mathcal{H}} \{L(H) + L(D|H)\}$$

- Note the resemblance with penalised model fitting, such as LASSO:
  - with LASSO we apply cross-validation to minimise

$$\arg\min_{\beta} \left\{ \frac{1}{n} \|\boldsymbol{y} - \boldsymbol{X}\beta\| + \lambda \|\beta\| \right\}$$

with the data part $\frac{1}{n}\|\boldsymbol{y} - \boldsymbol{X}\beta\|$ ($L(D|H)$ in MDL) and $\lambda\|\beta\|$ the model complexity part ($L(H)$ in MDL).

Example of dealing with overfitting (continued)



Questions to be answered

- The coding system to put down the description mechanism should not matter. How can we choose a universal programming language?
  - For example, whether we use `Matlab`, `R` or whatever programming language should not matter for the number of bits at use.
- How does MDL work in practice?
- What is the best approach: frequentist statistics, Bayesian statistics, machine learning, or MDL?
- What are good encoding systems?
- What patterns are generalisable and what not?
- How to incorporate prior knowledge?

Kolmogorov complexity

- Any application of MDL should not be affected by the programming language of choice.
- Andrey Kolmogorov postulated the invariance theorem:

  given any description of an object in a description language A, said description may be used in the optimal description language B with a constant overhead

- The description follows two steps:
  1. describe optimal language B in language A;
  2. describe object in optimal language B via language A.
- Comparable with a user-friendly programming language A (e.g. Matlab, R, Python) and whose code is compiled in an efficient programming language B (e.g. C++, C, FORTRAN).
- Part 1. is overhead independent of the object to be described, hence as the object size grows the overhead becomes small.

Kolmogorov complexity (example)

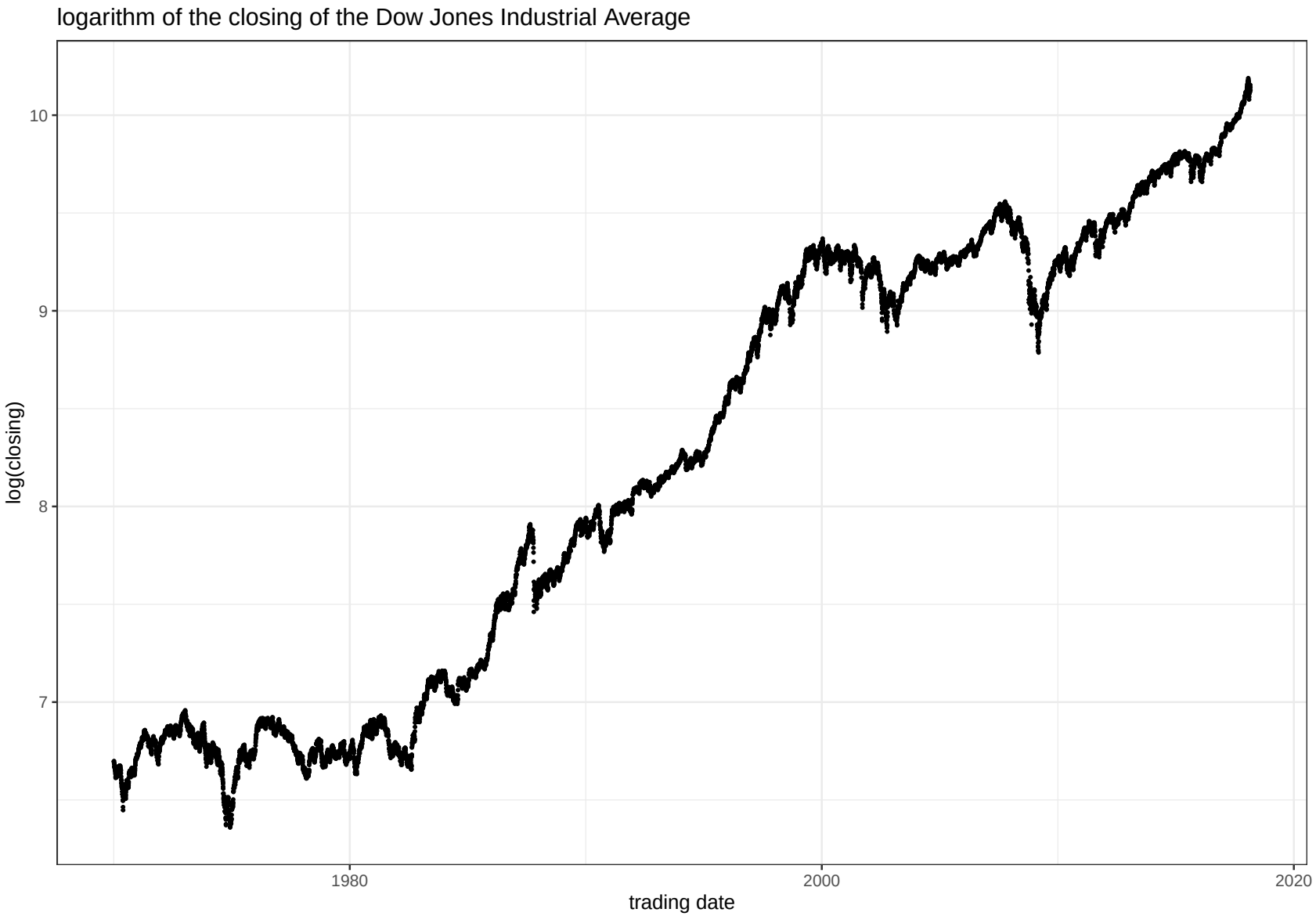
- Suppose we would like to describe the binary data sequence `1 0 0 0` in C via Matlab. How many bits does it take?
- First translate any code from C to Matlab needed to describe `1 0 0 0`.
  - E.g. C has a certain mapping procedure from the data type bit to its memory.
- The corresponding number of bits of the Matlab-code is $L_{\mathrm{C} \mapsto \mathrm{M}}$, e.g. 10k bits.
- To describe $D = \texttt{1 0 0 0}$ in C via Matlab takes $L_{\mathrm{C} \mapsto \mathrm{M}} + L_{\mathrm{M}}(D)$ bits.
- But $L_{\mathrm{C} \mapsto \mathrm{M}}$ is independent of the object to be described.
  - Whether we put in the works of Shakespeare or `Hello, World`, $L_{\mathrm{C} \mapsto \mathrm{M}}$ remains the same.

MDL in Practice

Dow Jones Industrial Average

- How can we employ MDL to compress a dataset?
  - Example taken from (Hansen and Yu 2001).
- Take the log daily return R_t and the volatility V_t of the Dow Jones Industrial Average (DJIA).
  - t runs from $t_0 =$ July 1962 until $T =$ June 1988, i.e. 6,430 trading days.
  - $R_t = P_t - P_{t-1}$ with P_t the logarithm of the DJIA at day t.
  - $V_t = 0.9V_{t-1} + 0.1R_t^2$ and V_0 the variance of the series P_t.
- R_t has a corresponding indicator: 1 (0) when $R_t > R_{t-1}$ ($R_t < R_{t-1}$).
  - Analogously for V_t.
  - Two binary strings of length 6,430 - 1 = 6,429.
  - R_t has 3,181 (49.49%) ups.
  - V_t has 2,023 or (31.47%) ups.

Dow Jones Industrial Average



Dow Jones Industrial Average

- Without data compression we require 6,429 bits per series.
- Using MDL we save 10% on the volatility series V_t while gaining nothing on P_t.
- n is known in advance so costs us $\lceil \log_2 n \rceil$ bits.
- Model the up or down indicator as a Bernoulli probability distribution with probability p on success.
  - Maximum likelihood yields $\hat{p} = k/n$ with k the number of successes.
- It takes us $-log_2(k/n)$ bits to model a success and $-log_2(1 - k/n)$ bits to model a failure.
- In total we need, ignoring rounding errors,

$$\log_2(n) + k\left[-log_2\left(\frac{k}{n}\right)\right] + (n-k)\left[-log_2\left(1-\frac{k}{n}\right)\right]$$

Dow Jones Industrial Average

- With $n = 6,429$ and
  - for R_t we have $k = 3181$ and $\hat{p} = \frac{k}{n} = \frac{3181}{6429} = 0.49$ we need

    $$\log_2(6429) + 3181[-\log_2 0.49] + 3248[-\log_2 0.51] = 6442 \text{ bits}$$

  - for V_t we have $k = 2023$ and $\hat{p} = \frac{k}{n} = \frac{2023}{6429} = 0.31$ we need

    $$\log_2(6429) + 2023[-\log_2 0.31] + 4406[-\log_2 0.69] = 5789 \text{ bits}$$

- In conclusion, for R_t we have gained nothing, even lost some bits, but the improvement for V_t is substantial.
  - The savings in storage increases as well as n increases or k/n approaches 0 or 1.
- In general, data compression leads to good model performance.
  - Note that data compression is not the main goal.

Approaches compared

How do the approaches compare: Frequentist

- Also known as orthodox or non-Bayesian.
- Main goal: models should be consistent.
  - At least asymptotically: as $n \to \infty$ the estimated model coefficients should converge to the 'true' model parameters.
  - In the example above, one would expect $\hat{\beta} = (5\,0\,2\,1)$ for $n \to \infty$.
- When the model assumptions are fulfilled, this approach yields the best possible predictions.
- Pros: widely known and applicable, easy calculation, models are good with infinite amount of data.
- Cons: assumptions are not realistic (e.g. almost no residual follows a normal probability distribution).

How do the approaches compare: Bayesian

- Main goal: incorporate prior knowledge by assigning a prior probability distribution.
  - Once the prior probability distribution is assigned, the posterior probability distribution can be derived / calculated.
  - Frequentist models consider the data random and the parameters fixed.
  - Bayesian models consider the data and the parameters random.
  - In practice, asymptotically frequentist and Bayesian models converge.
- Pros: mathematically elegant and precise, encompasses the frequentist approach.
- Cons: hard to come up with a useful prior probability distribution.

How do the approaches compare: Machine Learning

- Main goal: find and fit a model that is able to generalise from train to test data.
  - No assumptions about the data generating process.
- Pros: no assumptions required, good model performance.
- Cons: hard to explain predictions, hard to study the data generating process.

How do the approaches compare: MDL

- Main goal: infer useful information from the data and use that to achieve good data compression.
  - Good data compression implies good learning.
  - No assumptions about the data generating process required but are allowed.
- Pros: no assumptions required, has best of all worlds (Bayesian and Machine Learning): employ prior knowledge when viable, good statistical properties, and decent performance.
- Cons: relatively new in literature, intersection of state-of-the-art knowledge of information theory, measure theory and statistics.

Discussion

Application to Expected Credit Loss

- Suppose we apply the MDL principle on the calculation of the Expected Credit Loss.
  - No coding scheme discussed here, just the principle.
- Discussion:

  take a moving average of the realised credit losses to calculate the Expected Credit Loss
- Pros:
  - Occam's Razor: simpler is better.
  - Saves us a complete IFRS team to perform tedious calculations.
- Cons:
  - No new developments (e.g. Corona) taken into consideration. Can we with our current models?
  - No intermediate analyses available (e.g. use PD to rank customers in need).

Conclusion

Summary

- MDL is a guidance for model selection, an instrument against overfitting and exploratory data analysis.
- Sound statistics and decent model performance could make MDL a challenger for other statistical approaches.
- Much more to be explored.

What's next?

- A workshop to take a deepdive into the applications of MDL, backed up by statistical theory.
- Take a look at the open questions raised in this presentation.
  - What are good encoding systems?
  - What patterns are generalisable and what not?
  - How to incorporate prior knowledge?
- Dedicated application to IFRS and Regulatory Capital calculations.
- Please let Eelko (eelko.ubels@devolksbank.nl) or me (paul.vanleeuwen@devolksbank.nl) know whether you would like to join!

Questions

?

Bibliography

Grünwald, Peter D. 2007. *The Minimum Description Length Principle*. MIT press.

Hansen, Mark H, and Bin Yu. 2001. "Model Selection and the Principle of Minimum Description Length." *Journal of the American Statistical Association* 96 (454): 746–74.